Dinosaur Recalibration: Methods and Results

A Technical Companion to the Interactive Visualizations

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Abstract

We fit power ratings to ∼4,494 crowd-sourced animal combat matchups using a Bradley–Terry model, classify each animal by geological era, and apply a principled rescaling that corrects for what we argue is a systematic weight-based bias in human intuition about Mesozoic combat. Before fitting, a graph-connectivity preprocessing step removes poorly connected animals and injects structural constraints (gender geometric means, group-to-solo scaling, subspecies linking) that reduce noise without distorting real signal. The rescaling is anchored by a single constraint: that a *Dakotaraptor* overpowers a lion by the same multiplicative factor as a lion overpowers an unarmed human. We then fit weight-versus-score power laws via median regression for theropod dinosaurs and mammals separately, revealing that theropods carry a ∼9× built-in combat advantage at 1 kg—an advantage that grows to ∼14× at the smallest sizes, precisely where Mesozoic mammals lived. All data, code, and interactive visualizations are publicly available.

Contents

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1 Introduction

When people compare dinosaurs to modern animals, they tend to anchor on body mass. A *Tyrannosaurus rex* weighed roughly as much as an African elephant, so the instinct is to place them in the same tier. This paper argues that this reflects a **weight-based bias**—a default heuristic inherited from our experience with Cenozoic fauna, where mass is indeed the dominant predictor of combat outcome.

The Mesozoic operated under different rules. Theropod dinosaurs possessed body plans refined over 160 million years for predation: sickle claws, serrated teeth lining lever-arm jaws, bipedal agility, and attack postures that struck from above. These are *anatomical* advantages that scale differently from mass. We propose a correction factor derived from crowd-sourced combat data and a single calibration constraint, and we show that the corrected rankings produce a coherent picture of Mesozoic ecology.

This work serves as a technical companion to the interactive visualizations at `https://github.com/brunchwnewton/aethic-code/tree/main/nexic` and is motivated by the Dark Era hypothesis developed in the Nexic reasoning framework [1].

2 Data

2.1 Source

The raw data consists of pairwise animal combat matchups scraped from the Carnivora.net Interspecific Conflict Directory [4] Each matchup records two animals, a win probability for the first animal (as voted by site users), and the number of votes. After scraping, we obtain ∼4,494 matchup entries spanning ∼2,369 unique animals.

2.2 Unicode normalization

The source data contains corrupted Unicode characters (U+FFFD replacement characters from encoding errors). We apply an allowlist-based cleaning procedure: retain only ASCII letters, digits, and common punctuation; replace corrupted characters with a space at word boundaries and delete them within words. A manual corrections dictionary handles the remaining ambiguous cases.

2.3 Filtering

Matchups with win probabilities below 5% or above 95% are excluded as likely noise (one-sided stomps where voters are guessing rather than deliberating). This removes the most extreme outliers while preserving the informative middle range.

3 Graph Preprocessing

The raw matchup graph exhibits pathological connectivity issues. Many animals appear only in matchups with 0% or 100% win probability ("dead edges" that carry no information for Bradley–Terry), and numerous small clusters of animals are completely disconnected from the main component. For example, the House Mouse's only non-trivial matchups are against the Etruscan Shrew and Common Quail—a five-animal island whose BT scores float uncalibrated, inflating the House Mouse to 300× its correct value.

We apply a fully automated preprocessing pipeline with five layers of structural constraints, each designed to inject known biological relationships as soft priors that the BT model can use without overriding genuine signal.

3.1 Graph cleanup

We exclude three classes of poorly-connected animals:

1. Animals whose *only* matchups have 0% or 100% win probability (283 animals).
2. Animals on two-animal islands in the "real" graph (228 animals across 114 pairs).
3. Animals on larger islands with no plausible bridge to the main component (39 animals across 11 isolated ecosystem clusters: jellyfish, dragonflies, ants, etc.).

This removes 550 animals. The remaining 1,819 animals and 4,087 real matchups form the basis for structural augmentation.

3.2 Gender constraints

For animals with male, female, and base entries in the dataset (e.g. "Cheetah," "Cheetah (male)," "Cheetah (female)"), we add synthetic matchups that enforce a **geometric mean** constraint: the base entry's BT score should equal $\sqrt{\pi_{\text{male}} \cdot \pi_{\text{female}}}$.

This is achieved by adding two matchups at identical win probability p: "male beats base at p" and "base beats female at p." For any value of p, these two constraints force the BT solution to place the base at the geometric mean.[1] We use $p = 0.60$ at weight 10, yielding 55 gender constraints across 7 full trios and 41 pairs.

3.3 Group-to-solo scaling

Entries like "Grey Wolf (pack of 5)" are linked to "Grey Wolf" via synthetic matchups with win probability $P = N^{0.7}/(N^{0.7}+1)$, where N is the group size. The exponent 0.7 reflects coordination losses in group combat. This produces 270 group-to-solo links at weight 3.

3.4 Island bridges

For the remaining disconnected clusters, we add 12 manual bridge matchups at weight 3, each connecting an island animal to a similar animal in the main component with a domain-knowledge win probability (e.g. House Mouse $\leftrightarrow$ Brown Rat at 20%).

3.5 Subspecies linking

Animals entered under both a generic and a regional name (e.g. "Lion" and "African Lion") receive a 50% win-probability link at weight 20. Same-species subspecies (e.g. "Tiger" $\leftrightarrow$ "Siberian Tiger") receive size-adjusted links at weight 10. This pools matchup information across 51 subspecies pairs, reducing noise without forcing convergence:

[1] Proof: if $P(\text{male beats base}) = p$, then $\pi_{\text{base}} = \pi_{\text{male}} \cdot (1-p)/p$. If $P(\text{base beats female}) = p$, then $\pi_{\text{female}} = \pi_{\text{base}} \cdot (1-p)/p$. Multiplying: $\pi_{\text{male}} \cdot \pi_{\text{female}} = \pi_{\text{base}}^2$.

Pair	Before	After	Fix
Lion / African Lion	1.65×	1.18×	73%
Tiger / Bengal Tiger	3.01×	1.82×	59%
Am. Black Bear / Florida BB	4.85×	1.37×	90%

Table 1: Score ratio between same-species pairs, before and after subspecies linking.

3.6 Summary

The preprocessing adds 388 synthetic matchups to 4,087 real ones, producing a single connected graph of 1,819 animals. Synthetic matchups use tiered weights (3 for bridges/groups, 10 for gender/subspecies, 20 for same-animal duplicates) to remain subordinate to the real data. House Mouse drops from a score of 1,021 (Cheetah = 100 scale) to 3.3—a 300× correction—while well-connected animals shift by less than 5%.

4 Bradley–Terry Model

We assign each animal i a latent strength parameter $\pi_i > 0$ such that the probability of i defeating j is:

$$P(i \text{ beats } j) = \frac{\pi_i}{\pi_i + \pi_j} \tag{1}$$

Working in log-space ($\theta_i = \log \pi_i$), the log-likelihood of the observed data is:

$$\ell(\boldsymbol{\theta}) = \sum_{(i,j)} \left[w_{ij} \log \sigma(\theta_i - \theta_j) + w_{ji} \log \sigma(\theta_j - \theta_i) \right] - \frac{\lambda}{2} \|\boldsymbol{\theta}\|^2 \tag{2}$$

where $\sigma(x) = 1/(1 + e^{-x})$, w_{ij} is the implied win count for i over j, and $\lambda = 0.01$ provides L_2 regularization. We optimize (2) using L-BFGS-B.

5 Taxonomy Classification

Each animal is classified into one of three categories:

1. **Non-Mesozoic**: extant animals, Cenozoic/Paleozoic extinct species, synapsids, and Mesozoic mammals.
2. **Dinosaur**: non-avian Dinosauria from the Mesozoic.
3. **Mesozoic-other**: pterosaurs, marine reptiles, Mesozoic crocodylomorphs, and other non-dinosaurian Mesozoic archosaurs.

Classification uses manual overrides ($\sim$400 entries), name-based pattern matching, and a Wikipedia API fallback. Final classification: 154 dinosaurs, 55 mesozoic-other, 1,603 non-mesozoic.

6 Anatomy Correction

6.1 Lion anchor

We normalize all scores so that Lion = 100. The subspecies linking in preprocessing (§3) ensures that Lion, African Lion, Barbary Lion, etc. share information through the BT model, making a single entry sufficient as the anchor.

6.2 Human baseline

The unarmed human score is taken directly from the "Human" entry, which the gender constraints have already noise-reduced toward the geometric mean of "Human (male)" and "Human (female)":

$$\text{Human} = 5.15 \quad (\text{Lion} = 100) \tag{3}$$

6.3 The calibration constraint

The anatomy correction is defined by a single constraint:

$$\frac{\text{Dakotaraptor}_{\text{rescaled}}}{\text{Lion}} = \frac{\text{Lion}}{\text{Human}} \tag{4}$$

In log-space, Dakotaraptor, Lion, and Human are equally spaced. Since Lion = 100 and Human = 5.15:

$$\text{Dakotaraptor}_{\text{rescaled}} = \frac{100^2}{5.15} = 1{,}940 \tag{5}$$

The dinosaur scale factor is $f_{\text{dino}} = 1{,}940/191.3 \approx 10.14$. The mesozoic-other factor is $\sqrt{f_{\text{dino}}} \approx 3.18$.

6.4 Motivation

The lion-versus-unarmed-human comparison is the point at which anatomical difference is most viscerally intuitive: a lion's claws, jaws, and reflexes make the outcome near-instant, not because the lion is heavier, but because its anatomy is purpose-built weaponry. The thesis is that a *Dakotaraptor*'s jaw, sickle claws, height, and agility produce the same kind of mismatch against a lion.

6.5 Results

Animal	Crowd	Rescaled	Factor	Category
Tyrannosaurus rex	10,584	107,344	10.14×	dinosaur
Dakotaraptor steini	191	1,940	10.14×	dinosaur
Velociraptor mongoliensis	1.7	16.8	10.14×	dinosaur
Mosasaurus hoffmannii	5,525	17,597	3.18×	mesozoic-other
Lion	100	100	1.00×	non-mesozoic
Tiger	148	148	1.00×	non-mesozoic
Human	5.2	5.2	1.00×	non-mesozoic

Table 2: Selected animals before and after anatomy correction.

7 Weight–Score Analysis

7.1 Median regression

We fit power laws of the form score $= C \cdot w^b$ in log-space using **median regression**: minimize $\sum_i |\log s_i - a - b \log w_i|$. This is robust to crowd-data outliers while preserving the central tendency. Mammals below 1 kg are excluded from the regression: BT scores for tiny mammals are relative within their matchup pool and do not follow the weight-based power law at extreme low weights.

7.2 Results

The fitted power laws are:

$$\text{Theropods:} \quad s = 0.549 \cdot w^{1.263} \quad (\text{pseudo-}R^2 = 0.56,\ n = 26) \tag{6}$$

$$\text{Mammals } (\geq 1\,\text{kg})\text{:} \quad s = 0.060 \cdot w^{1.376} \quad (\text{pseudo-}R^2 = 0.68,\ n = 35) \tag{7}$$

The theropod intercept is ~9× the mammal intercept, representing a built-in body-plan advantage at every mass. The mammal exponent is steeper (1.376 vs. 1.263), meaning mammals narrow the gap with increasing size, but from such a deficit that they never overtake within biologically relevant ranges.

Weight (kg)	Theropod	Mammal	Ratio	Context
0.02	< 0.01	< 0.01	14×	Mouse-sized
0.1	0.03	< 0.01	12×	Rat-sized (our ancestor)
2	1.3	0.2	9×	Squirrel-sized
15	16.8	2.5	7×	Fox-sized (*Velociraptor*)
75	128	23	6×	Human-sized (*Deinonychus*)
250	587	119	5×	Lion-sized (*Dakotaraptor*)
500	1,408	309	5×	Bear-sized (*Utahraptor*)

Table 3: Predicted combat scores at equal body mass. The anatomy advantage is largest at the smallest sizes.

7.3 Dark Era implications

A 100-gram rat-sized Mesozoic mammal (our ancestor) has a predicted score of ~0.003. The smallest known theropods (e.g. *Yi qi* at ~400 g) represent a ~69× mismatch. A 3 kg *Compsognathus*-class hunter: ~870×. A 15 kg *Velociraptor*: ~6,700×.

These ratios quantify the ecological pressure described by the Dark Era hypothesis [1]: our mammalian ancestors survived not by competing with theropods but by hiding from them—nocturnal, burrowing, small. The entire mammalian cognitive architecture, from the limbic system to the structure of our senses, was forged under this predation pressure.

8 Discussion

Several caveats apply. The crowd-sourced data reflects voter knowledge and biases, not controlled experiments. The Bradley–Terry model assumes transitivity and context-independence, which real combat violates. The anatomy correction factor of ~10× is derived from a single calibration constraint, not from biomechanical first principles. And the weight estimates for extinct species carry substantial uncertainty.

Despite these limitations, the analysis reveals a coherent pattern: once the weight-based bias is corrected, the rescaled rankings produce an ecologically sensible picture in which theropods dominate at every size class, herbivorous dinosaurs outclass the largest Cenozoic land mammals, and the smallest Mesozoic theropods are already formidable predators of mammal-sized prey. The consistency of this picture across four orders of magnitude of body mass suggests that the correction is capturing something real about the difference between Mesozoic and Cenozoic combat ecology.

9 Code and Data Availability

All code, data, and interactive visualizations are available at:

https://github.com/brunchwnewton/aethic-code/tree/main/nexic/data

The pipeline consists of seven scripts:

1. `1_link_scraper.py` — Scrapes matchup URLs from the directory
2. `2_scrape_duels.py` — Extracts pairwise matchup data from each URL
3. `3_preprocess_duels.py` — Graph cleanup and structural constraints
4. `4_power_assembler.py` — Bradley–Terry model fitting
5. `5_categorizer.py` — Taxonomy classification
6. `6_rescale_dinos.py` — Anatomy correction and rescaling
7. `7_curve_fit.py` — Weight–score analysis and median regression

Matchup data sourced from the Carnivora.net Interspecific Conflict Directory [4]

A Theropod Weight Estimates

Table 4 lists the theropod data points used in the weight–score power law regression (§7), sorted by body mass. Weight estimates are drawn from paleontological literature and represent adult averages. Rescaled scores reflect the anatomy-corrected values (Lion = 100).

Theropod	Clade	Weight (kg)	Score
Yi qi	Scansoriopterygid	0.4	1.3
Archaeopteryx lithographica	Basal avialan	0.9	0.9
Compsognathus longipes	Compsognathid	3	0.9
Ornitholestes hermanni	Maniraptoran	12	2.8
Linheraptor exquisitus	Dromaeosaurid	12	3.9
Velociraptor mongoliensis	Dromaeosaurid	15	16.8
Dromaeosaurus albertensis	Dromaeosaurid	15	16.4
Coelophysis bauri	Coelophysid	20	5.0
Suskityrannus hazelae	Tyrannosauroid	30	178.2
Stenonychosaurus inequalis	Troodontid	50	45.5
Deinonychus antirrhopus	Dromaeosaurid	73	305.3
Citipati osmolskae	Oviraptorid	75	13.1
Moros intrepidus	Tyrannosauroid	78	88.2
Liliensternus liliensterni	Coelophysoid	130	110.4
Timurlengia euotica	Tyrannosauroid	170	360.4
Ornithomimus edmontonicus	Ornithomimid	170	12.5
Marshosaurus bicentesimus	Megalosauroid	200	877.6
Herrerasaurus ischigualastensis	Herrerasaurid	210	2,529
Dakotaraptor steini	Dromaeosaurid	250	1,940
Achillobator giganticus	Dromaeosaurid	250	1,277
Dilophosaurus wetherilli	Dilophosaurid	400	1,186
Gallimimus bullatus	Ornithomimid	440	662
Utahraptor ostrommaysorum	Dromaeosaurid	500	2,705

Table 4: Theropod data points for the weight–score regression.

B Interleaved Rankings: Mammals and Dinosaurs

Table 5 presents a curated selection of 133 animals from across the rescaled rankings, sorted *descending* by combat score with Lion = 100. Dinosaurs are marked with ▲ and Mesozoic non-dinosaurs with △; unmarked entries are non-Mesozoic. Solo entries only—groups, packs, and female-specific entries are excluded. Weights are adult averages from the paleontological and zoological literature. Familiar reference animals (dog breeds, farm animals, common wildlife) are interspersed throughout as intuitive scale markers.

The interleaving demonstrates the central claim: after anatomy correction, theropods of a given mass consistently outrank mammals several times their weight. A 73 kg *Deinonychus* ties a 450 kg Polar Bear at 305 (rows 32–33). A 250 kg *Dakotaraptor* outscores a 6-tonne Woolly Mammoth (rows 16–17). At the small end, 15 kg dromaeosaurids rank alongside 55–68 kg mammalian predators (rows 62–65). And a 30 kg *Suskityrannus*—a juvenile tyrannosaur the size of a Kangal—scores higher than a Tiger (rows 43 vs. 48).

#	Score	Animal	Weight[2]	
1	114,463	*Ankylosaurus magniventris*	6.0 t	▲
2	107,344	*Tyrannosaurus rex*	8.4 t	▲
3	86,822	*Giganotosaurus carolinii*	6.5 t	▲
4	81,449	*Triceratops horridus*	9.0 t	▲
5	54,927	*Diplodocus carnegii*	13 t	▲
6	35,008	*Euoplocephalus tutus*	2.5 t	▲
7	33,036	*Brontosaurus excelsus*	15 t	▲
8	29,645	*Spinosaurus aegyptiacus*	7.0 t	▲
9	28,365	*Stegosaurus stenops*	3.5 t	▲
10	22,221	*Daspletosaurus torosus*	2.5 t	▲
11	19,393	*Styracosaurus albertensis*	2.7 t	▲
12	17,597	*Mosasaurus hoffmannii*	5.0 t	△
13	17,285	*Tarbosaurus bataar*	5.0 t	▲
14	14,865	*Allosaurus fragilis*	2.0 t	▲
15	7,525	*Parasaurolophus spp.*	2.5 t	▲
16	4,204	*Iguanodon bernissartensis*	3.5 t	▲
17	3,894	*Edmontosaurus annectens*	4 t	▲
18	3,883	*Carnotaurus sastrei*	1.5 t	▲
19	2,981	Columbian Mammoth	10.0 t	
20	2,705	*Utahraptor ostrommaysorum*	500 kg	▲
21	2,529	*Herrerasaurus ischigualastensis*	210 kg	▲
22	2,205	Orca (Killer Whale)	5.4 t	
23	1,940	*Dakotaraptor steini*	250 kg	▲
24	1,447	Woolly Mammoth	6.0 t	
25	1,446	Woolly Rhinoceros	2.5 t	
26	1,277	*Achillobator giganticus*	250 kg	▲
27	1,274	Megatherium americanum	4.0 t	
28	1,186	*Dilophosaurus wetherilli*	400 kg	▲
29	985	African Bush Elephant	6.0 t	
30	882	White Rhinoceros	2.3 t	
31	878	*Marshosaurus bicentesimus*	200 kg	▲

#	Score	Animal	Weight[2]	
32	853	Great White Shark	1.1 t	
33	710	*Pachycephalosaurus wyomingensis*	450 kg	▲
34	662	*Gallimimus bullatus*	440 kg	▲
35	544	*Anzu wyliei*	200 kg	▲
36	459	Arctodus simus	800 kg	
37	430	Hippopotamus	1.5 t	
38	417	Saltwater Crocodile	450 kg	
39	360	*Timurlengia euotica*	170 kg	▲
40	305	Polar Bear	450 kg	
41	305	*Deinonychus antirrhopus*	73 kg	▲
42	274	Smilodon populator	400 kg	
43	235	American Bison	900 kg	
44	224	*Quetzalcoatlus northropi*	250 kg	△
45	212	Kodiak Bear	600 kg	
46	203	Smilodon fatalis	220 kg	
47	178	*Suskityrannus hazelae*	30 kg	▲
48	176	(Wild) Water Buffalo	900 kg	
49	161	Water Buffalo	800 kg	
50	160	Wood Bison	900 kg	
51	150	American Alligator	230 kg	
52	148	Tiger	220 kg	
53	133	Grizzly Bear	270 kg	
54	110	*Liliensternus liliensterni*	130 kg	▲
55	109	Alaskan Moose	600 kg	
56	**100**	**Lion**	**190 kg**	
57	88	Siberian Tiger	230 kg	
58	88	*Moros intrepidus*	78 kg	▲
59	85	African Lion	185 kg	
60	81	Bengal Tiger	220 kg	
61	48	Jaguar	96 kg	
62	46	*Stenonychosaurus inequalis*	50 kg	▲
63	38	*Protoceratops andrewsi*	180 kg	▲
64	37	Western Gorilla	160 kg	
65	31	Cougar	70 kg	
66	30	Leopard	60 kg	
67	24	Eastern Gorilla	160 kg	
68	23	Komodo Dragon	70 kg	
69	20	Green Anaconda	30 kg	
70	18	Wild Boar	80 kg	
71	17	Spotted Hyena	55 kg	
72	17	Dire Wolf	68 kg	
73	17	*Velociraptor mongoliensis*	15 kg	▲
74	16	*Dromaeosaurus albertensis*	15 kg	▲
75	13	*Citipati osmolskae*	75 kg	▲
76	13	Elk	330 kg	
77	13	*Ornithomimus edmontonicus*	170 kg	▲

#	Score	Animal	Weight[2]	
78	10	Common Warthog	80 kg	
79	9.7	Grey Wolf	40 kg	
80	7.9	Burmese Python	90 kg	
81	7.5	Neanderthal	78 kg	
82	6.4	Bornean Orangutan	75 kg	
83	5.9	Human (male)	80 kg	
84	5.9	Neapolitan Mastiff	60 kg	
85	5.8	Ostrich	110 kg	
86	5.8	*Psittacosaurus lujiatunensis*	20 kg	▲
87	5.7	Dogo Argentino	43 kg	
88	5.2	Human	70 kg	
89	5.0	*Coelophysis bauri*	20 kg	▲
90	4.8	Caucasian Shepherd Dog	55 kg	
91	4.5	*Pteranodon longiceps*	25 kg	△
92	4.5	Rottweiler	50 kg	
93	4.4	Cheetah	50 kg	
94	4.3	Tibetan Mastiff	55 kg	
95	4.2	Red Kangaroo	85 kg	
96	3.9	*Linheraptor exquisitus*	12 kg	▲
97	3.9	Wolverine	14 kg	
98	3.5	Mandrill	25 kg	
99	3.3	Common Chimpanzee	50 kg	
100	3.1	Kangal	55 kg	
101	2.9	Southern Cassowary	60 kg	
102	2.8	*Ornitholestes hermanni*	12 kg	▲
103	2.5	Boerboel	65 kg	
104	2.2	Thylacine	25 kg	
105	2.1	German Shepherd	34 kg	
106	1.8	Human (female)	55 kg	
107	1.8	Ocelot	12 kg	
108	1.7	King Cobra	6 kg	
109	1.7	American Pit Bull Terrier	27 kg	
110	1.7	Honey Badger	11 kg	
111	1.5	Bonobo (Pygmy Chimpanzee)	40 kg	
112	1.5	Caracal	13 kg	
113	1.4	Fishing Cat	10 kg	
114	1.3	Bobcat	9 kg	
115	1.3	*Yi qi*	400 g	▲
116	1.2	Coyote	14 kg	
117	1.1	Labrador Retriever	30 kg	
118	1.0	Savannah Cat	8 kg	
119	0.9	*Archaeopteryx lithographica*	900 g	▲
120	0.9	*Compsognathus longipes*	3 kg	▲
121	0.7	Siberian Husky	23 kg	
122	0.6	Serval	12 kg	
123	0.5	Fisher	4.5 kg	

#	Score	Animal	Weight[2]
124	0.5	Brown (Norway) Rat	300 g
125	0.4	Emu	40 kg
126	0.4	Patterdale Terrier	6 kg
127	0.3	Wild Turkey	8 kg
128	0.2	Red Fox	6 kg
129	0.2	Maine Coon Cat	5 kg
130	0.1	House Mouse	20 g
131	0.1	Fennec Fox	1 kg
132	0.1	Feral Cat	4 kg
133	0.1	Etruscan Shrew	2 g

Table 5: Interleaved combat rankings, descending (Lion = 100, row 52). ▲ = dinosaur, △ = Mesozoic non-dinosaur. Dog breeds, farm animals, and common wildlife are interspersed as intuitive scale markers. Key juxtapositions: rows 32–33 (73 kg *Deinonychus* = 450 kg Polar Bear), rows 16–17 (250 kg *Dakotaraptor* > 6 t Woolly Mammoth), rows 68–71 (15 kg raptors ≈ 55–68 kg wolf-tier predators).

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